

04-02-26
Wednesday

Module - 2

Discrete Probability distributions

Random Experiment: A experiment in which the result cannot be predicted or we cannot know. Random Experiment is an Experiment which has more one possible no. of outcome

eg: (1) Tossing a coin

(2) Tossing a two coin

Samplespace: The set of all outcomes of a random Experiment

eg: (1) Tossing a coin

samplespace $S = \{H, T\}$

(2) Tossing 2 coin

$S = \{HH, HT, TH, TT\}$

Random variable: Just assigning a numeric value to sample space element. A Random variable 'X' is a function from Samplespace 'S' to Real number 'R'.

eg: Tossing a coin

$S = \{H, T\}$

X: no of heads

$X(H) = 1$

$X(T) = 0$

$\Rightarrow X = 0, 1$

In case of continuous
like life time of a bulb;
 $X = 0 \rightarrow \infty$

Probability mass function

A function $p(x)$ from a random variable ' x ' to real numbers is called probability mass function if it satisfies two conditions:

1. $p(x) \geq 0$

$$[p(x) = P(X=x)]$$

2. $\sum_x p(x) = 1$

Eg: Tossing a coin:

$$S = \{H, T\}$$

X = no. of heads

$$X(H) = 1$$

$$X(T) = 0$$

X	0	1
$p(x)$	$\frac{1}{2}$	$\frac{1}{2}$

$$\Rightarrow p(x) \geq 0$$
$$\sum_x p(x) = 1$$

Tossing 2 coin:

$$S = \{HH, HT, TH, TT\}$$

X = no. of heads.

$$X \in \{0, 1, 2\}$$

X	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$p(x) \geq 0$$
$$\sum_x p(x) = 1$$

Q1 Determine whether the following can be probability mass function of a Random variable which takes the value: 1, 2, 3, 4 and

1) $* b(1) = 0.16$

$* b(2) = 0.28$

$* b(3) = 0.28$

$* b(4) = 0.28$

~~Q2~~ 2) $* b(1) = 0.37$

$b(2) = 0.31$

$b(3) = 0.34$

$b(4) = -0.02$

Q2 3) Check whether the following is pmf

1) $b(n) = \frac{n}{14}, n = 0, 1, 2, 3, 4$

2) $b(n) = \frac{3-x^2}{4} = 0, 1, 2$

Ans 1) here $b(1), b(2), b(3), b(4) > 0$

and $b(1) + b(2) + b(3) + b(4)$

$\Rightarrow 0.16 + 0.28 + 0.28 + 0.28$

$= 1$

$\therefore b(1) + b(2) + b(3) + b(4) = 1$

$\Rightarrow$ The given can be PMF.

2) here $b(1), b(2), b(3) > 0$

but $b(4) < 0$

So, the given cannot be PMF.

Ans Q2

$$1: f(n) = \frac{n}{14}; \quad n = 0, 1, 2, 3, 4$$

when $n=0$

$$f(n) \geq 0$$

when $n=1$

$$f(n) \geq 0 \Rightarrow f(1) = \frac{1}{14}$$

when $n=2$

$$f(n) \geq 0 \quad f(2) = \frac{2}{14}$$

when $n=3$

$$f(n) \geq 0 \quad f(3) = \frac{3}{14}$$

when $n=4$

$$f(n) \geq 0; \quad f(4) = \frac{4}{14}$$

Condition 1 satisfied

$$\sum_{n=0,1,2,3,4} f(n)$$

$$\Rightarrow 0 + \frac{1}{14} + \frac{2}{14} + \frac{3}{14} + \frac{4}{14} \neq 1$$

Condition 2 not satisfied.

$\Rightarrow$ So not a PMF.

$$2: f(x) = \frac{3-x^2}{4}, \quad x = 0, 1, 2.$$

when $x=0$

$$f(x) = \frac{3}{4} > 0$$

when $x=1$

$$f(x) = \frac{2}{4} > 0$$

when $n = 2$

$$b(x) = -\frac{1}{4} < 0$$

so condition 1 violated
~~so~~ $\therefore$ Not a PMF

Cumulative probability Distribution function

$$F(n) = P(X \leq n)$$

Eg: Tossing 2 coins

$X \rightarrow$ no. of head

$$x = 0, 1, 2$$

$$x = 0 \quad 1 \quad 2$$

$$P(x) = \frac{1}{4} \quad \frac{1}{2} \quad \frac{1}{4}$$

11.02.2026
Wednesday

Q1

A store carries flash drive with either 1 GB, 2 GB, 4 GB, 8 GB or 16 GB of memory. The accompanying table gives the distribution of Y = the amount of memory in purchase drive.

Y	1	2	4	8	16
$P(Y)$	0.05	0.10	0.35	0.40	0.10

Ans We need to find cumulative probability function $F(y)$

$$F(y) = P(Y \leq y)$$

$$F(1) = P(Y \leq 1) = P(1) = 0.05$$

$$F(2) = P(Y \leq 2) = P(1) + P(2) = 0.05 + 0.10 \\ = 0.15$$

$$F(4) = P(Y \leq 3) = P(1) + P(2) + P(3) = 0.05 + \\ 0.10 + 0.35 \\ = 0.50$$

$$F(8) = P(Y \leq 8) = P(1) + P(2) + P(3) + P(8) = \\ 0.05 + 0.10 + 0.35 + 0.40 \\ = 0.90$$

$$F(16) = P(Y \leq 16) = P(1) + P(2) + P(3) + P(8) \\ + P(16) \\ = 0.05 + 0.10 + 0.35 + 0.40 + \\ 0.10 \\ = 1.00$$

$$F(y) = \begin{cases} 0 & : y < 1 \\ 0.05 & : 1 \leq y < 2 \\ 0.15 & : 2 \leq y < 4 \\ 0.50 & : 4 \leq y < 8 \\ 0.90 & : 8 \leq y < 16 \\ 1 & : y \geq 16 \end{cases}$$

Q2. Let $b(n) = k/2^n$ for $x = 0, 1, 2, 3$ and 4 be a probability mass function of a random variable. Find k and distribution function $F(n)$

Ans $b(n) = \frac{k}{2^n}$, $n = 0, 1, 2, 3, 4$.

The sum of probabilities over all values of $n = 1$

$$\Rightarrow \sum_{n=0}^4 b(n) = 1.$$

$$\sum_{n=0}^4 \frac{k}{2^n} = k \left(\frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} \right)$$

$$1 = k \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \right)$$

$$= k \left(\frac{16 + 8 + 4 + 2 + 1}{16} \right)$$

$$= k \left(\frac{31}{16} \right)$$

$$k = \frac{16}{31} \therefore \text{COF} = \underline{\underline{\frac{16}{31}}}$$

$$F(n) = \frac{16}{31} \left(\frac{1}{2^n} \right) \text{ for all } n = 0, 1, 2, 3, 4$$

CDF:

$$F(0) = P(X \leq 0) = f(0) = \frac{16}{31} \times \frac{1}{2} = \frac{16}{31}$$

$$F(1) = P(X \leq 1) = f(0) + f(1)$$

$$= \frac{16}{31} + \frac{8}{31} = \underline{\underline{\frac{24}{31}}}$$

$$\begin{aligned}
 F(2) &= P(X \leq 2) = f(0) + f(1) + f(2) \\
 &= \frac{24}{31} + \frac{4}{31} \\
 &= \frac{28}{31}
 \end{aligned}$$

$$\begin{aligned}
 F(3) &= P(X \leq 3) = f(0) + f(1) + f(2) + f(3) \\
 &= \frac{28}{31} + \frac{2}{31} \\
 &= \frac{30}{31} \\
 &= \underline{\underline{\frac{30}{31}}}
 \end{aligned}$$

$$\begin{aligned}
 F(4) &= P(X \leq 4) = f(0) + f(1) + f(2) + f(3) + f(4) \\
 &= \frac{30}{31} + \frac{1}{31} \\
 &= \frac{31}{31} \\
 &= 1
 \end{aligned}$$

$$f(n) = \begin{cases} \frac{16}{31} & : n \leq 0 \\ \frac{16}{31} & : 0 \leq n < 1 \\ \frac{24}{31} & : 1 \leq n < 2 \\ \frac{28}{31} & : 2 \leq n < 3 \\ \frac{30}{31} & : 3 \leq n < 4 \\ 1 & : n \geq 4 \end{cases}$$

Mean and variance of a random variable.

The mean / expectation of random variable 'x' with PMF $p(x)$ is denoted by $E(x)$

$$E(x) = \sum_n x \cdot p(x)$$

$$E(x^2) = \sum_n x^2 \cdot p(x)$$

And variance of 'x', $V(x)$

$$V(x) = E(x^2) - [E(x)]^2$$

Q1. In a certain city during the month of July there will be 0, 1, 2 and 3 power failures with probabilities 0.4, 0.3, 0.2 and 0.1.

X = power failure in random city of month July

Find mean and median of x .

Ans:

PMF =

x	0	1	2	3
p(x)	0.4	0.3	0.2	0.1

Mean :

$E(x)$:

$$E(x) = 0 \times p(0) + 1 \times p(1) + 2 \times p(2) + 3 \times p(3)$$

$$= 0.04 + 1 \times 0.3 + 2 \times 0.2 + 3 \times 0.1$$

$$= 0.3 + 0.4 + 0.3 = \underline{1}$$

Variance

$$\begin{aligned} E(x^2) &= 0^2 * p(0) + 1^2 * p(1) + 2^2 * p(2) + \\ &\quad 3^2 * p(3) \\ &= 0.3 + 0.8 + 0.9 \\ &= \underline{\underline{2}} \end{aligned}$$

Variance:

$$\begin{aligned} V(x) &= E(x^2) - [E(x)]^2 \\ &= 2 - 1 \\ &= \underline{\underline{1}} \end{aligned}$$

13.02.26
Friday

Binomial Distribution.

Consider a random experiment with only two possible outcomes: success and failure (both are independent)

Let 'P' be probability of success +
'q' be probability of failure ($q = P - 1$).

Suppose the experiment conducted 'n' times, define a random variable 'X' such that it is the no. of success in 'n' trial. then the probability mass function

$$P(X=x) = {}^n C_x P^x q^{n-x} \quad ; \text{ where } x=0, 1, 2, \dots, n.$$

The random variable 'x' is called binomial random variable and the probability distribution is called binomial distribution and is denoted where X follows $B(n, p)$:

$$X \sim B(n, p).$$

Mean of X or Expectation of X : denoted by $E(x)$

$$E(x) = n * p$$

$$\text{Variance of } n \quad V(x) = n * p * q$$

Q. Find the binomial distribution for which mean is 5 and variance is $15/4$.

Ans

Given :

$$\text{mean: } E(x) = 5$$

$$\text{Variance: } V(x) = \frac{15}{4}$$

To find: $X \sim B(n, p)$

$$\Rightarrow E(x) = np = 5 \quad \text{--- (1)}$$

$$\Rightarrow V(n) = npq = \frac{15}{4} \quad \text{--- (2)}$$

$$(np)q = \frac{15}{4}$$

$$\cancel{5}q = \frac{\cancel{15}}{4} \quad \text{--- from 1}$$

$$\boxed{q = 3/4}$$

We know that it is a binomial distribution so only 2 possible outcomes p & q where

$$p + q = 1 \quad \text{--- (3)}$$

from (3)

$$p = 1 - q$$

$$p = 1 - \frac{3}{4}$$

$$\boxed{p = \frac{1}{4}}$$

$$\text{Now } E(X) = np = 5$$

$$\Rightarrow n \times \frac{1}{4} = 5$$

$$\Rightarrow \boxed{n = 20}$$

$$\therefore X \sim B(n, p)$$

$$\Rightarrow X \sim B(20, \frac{1}{4})$$

$$\Rightarrow P(X=0) = {}^{20}C_0 p^0 q^{20} = {}^{20}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{20}$$

$$P(X=1) = {}^{20}C_1 p^1 q^{19} = {}^{20}C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{19}$$

$$P(X=20) = {}^{20}C_{20} p^{20} q^0$$

$$= {}^{20}C_{20} \left(\frac{1}{4}\right)^{20} \left(\frac{3}{4}\right)^0$$

Q2 If the probability of a new born child is male in a typical family is 0.6. Find the probability that in a family of 5 children

1) There are exactly 3 boys

2) Majority of girls.

Ans Given :

$$P(\text{probability of male}) = 0.6$$

$$\Rightarrow q = 1 - p = 0.4$$

Let X = no. of males

$n = 5$ (total no. of children)

$$\Rightarrow X \sim B(5, 0.6)$$

1) exactly 3 boys $\Rightarrow x = 3$

$$P(X=3) = {}^5C_3 (0.6)^3 (0.4)^2$$

$$= \frac{5!}{(5-3)! \times 3!} \times (0.6)^3 \times (0.4)^2$$

$$= 10 \times 0.216 \times 0.16$$

$$= \underline{\underline{0.3456}}$$

$\therefore$ The probability of exactly 3 boys in a family is 0.3456.

2) Majority of girls

This implies that there should be

Boys (0, 1, 2) i.e. ~~either~~

$$\begin{aligned} P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= {}^5C_0 (0.6)^0 (0.4)^5 + {}^5C_1 (0.6)^1 (0.4)^4 \\ &\quad + {}^5C_2 (0.6)^2 (0.4)^3 \end{aligned}$$

$$= \frac{5!}{(5-0)!0!} \times (0.6)^0 \times (0.4)^5 + \frac{5!}{(5-1)!1!} \times (0.6)^1 \times (0.4)^4$$

$$+ \frac{5!}{(5-2)!2!} \times (0.6)^2 \times (0.4)^3$$

$$= 0.01024 + 0.0768 + 0.2304$$

$$= \underline{\underline{0.31744}}$$

$\therefore$ The probability of majority of new-born be girls is 0.31744

Q3 3111 dies are thrown 729 times how many times do you expect atleast 3 dies to show a '5 or 6'.

Ans Let 'n' = total no: of dies = 6
 N = total no: of trials = 729
 X = no: of times getting 5 or 6

For a single die

$$\begin{array}{ccc} \text{probability of getting 5} & = & \frac{1}{6} \\ \text{"} & & \text{"} \\ \text{"} & & \text{"} \\ \text{6} & = & \frac{1}{6} \end{array}$$

$$\therefore P = \text{probability of getting either 5 or 6}$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

$$P = \frac{1}{3} \Rightarrow q = 1 - P \Rightarrow 1 - \frac{1}{3} = \frac{2}{3}$$

Now we know for one die $P = \frac{1}{3}$
 $\therefore$ so for 6 die the prob showing 5 or 6 is
 $X \sim B(6, \frac{1}{3}) =$

$$P(X \geq 3) = P(3) + P(4) + P(5) + P(6)$$

$$\Rightarrow P(X=3) = {}^6C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 = \frac{160}{729}$$

$$\Rightarrow P(X=4) = {}^6C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 = \frac{60}{729}$$

$$\Rightarrow P(X=5) = {}^6C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^1 = \frac{12}{729}$$

$$\Rightarrow P(X=6) = {}^6C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^0 = \frac{1}{729}$$

$$\Rightarrow P(X \geq 3) = \frac{160}{729} + \frac{60}{729} + \frac{12}{729} + \frac{1}{729}$$

$$= \frac{233}{729}$$

$\therefore$ The probability of getting 5 or 6 when '6' dies are thrown one is $\frac{233}{729}$, So if it is repeated 729

time then the probability of getting 5 or 6 in 729 time of throwing 6 die is

$$= 729 \times \text{probability of throwing one}$$

$$= 729 \times \frac{233}{729} = \underline{\underline{233 \text{ times}}}$$

18.02.26
Wednesday

Poisson Distribution.

A poisson distribution is applicable only if the following conditions are satisfied.

1. The no: of trials n is very large
2. The probability of success ' p ' is very small. The product of n and p , $\lambda = np$ is a constant.

A discrete random variable ' x ' taking the value $0, 1, 2, \dots$, etc is said to follow Poisson distribution with parameter ' λ ' as PMF, $P(X) = \frac{e^{-\lambda} \lambda^n}{n!}$; $n = 0, 1, 2, 3, \dots$

eg: • Customer arriving in a call center.
• No: of typing mistakes in a book.

Mean and Variance of Poisson Distribution

Q. Derive Mean of poisson distribution

Ans. Given $P(X) = \frac{e^{-\lambda} \lambda^n}{n!}$; $n = 0, 1, 2, \dots$

In binomial

$$E(x) = \sum_{n=0,1,\dots} n \cdot p(n)$$

$$= \sum_{n=0,1,\dots} n \cdot \frac{e^{-\lambda} \lambda^n}{n!}$$

$$\boxed{n! = n(n-1)(n-2)\dots(2 \times 1) \cdot 1}$$

$$= \sum_{n=0,1,\dots} \cancel{n} \cdot \frac{e^{-\lambda} \lambda^n}{\cancel{n} (n-1)!}$$

$$= \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n-1)!}$$

$$\boxed{e^n = 1 + \frac{n}{1!} + \frac{n^2}{2!} + \frac{n^3}{3!} \dots = \sum_{n=0}^{\infty} \frac{n^n}{n!}} \quad \text{--- ①}$$

$$= e^{-\lambda} \sum_{n=1}^{\infty} \frac{\lambda^n}{(n-1)!}$$

$$\boxed{\lambda^n = \lambda \cdot \lambda^{n-1}}$$

$$= e^{-\lambda} \cdot \lambda \sum_{n=1}^{\infty} \frac{\lambda^{(n-1)}}{(n-1)!}$$

$$= e^{-\lambda} \cdot \lambda \cdot e^{\lambda} \quad \text{--- from ①}$$

$$= \underline{\underline{\lambda}}$$

H.W. Q. Derive Variance of poisson distribution.

Q

A random variable follows poisson distribution such that $P(X=0) = \frac{2}{3} P(X=1)$. Find $P(X=3)$

Ans: Given

$$P(X=0) = \frac{2}{3} P(X=1)$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^n}{n!}$$

$$\Rightarrow P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!}$$

$$= \underline{e^{-\lambda}} \quad \text{--- ①}$$

compare ① with RHS.

$$e^{-\lambda} = \frac{2}{3} \frac{e^{-\lambda} \lambda^1}{1!}$$

$$1 = \frac{2}{3} \lambda$$

$$\underline{\underline{\lambda = \frac{3}{2}}}$$

$$\Rightarrow P(X=x) = \frac{e^{(-3/2)} (3/2)^n}{n!}$$

when $n=3$

$$= \frac{e^{(-3/2)} (3/2)^3}{3!}$$

Q.

IF 3% of electrical bulb manufactured by a company is defective. Find the probability that in a sample of 100 bulb there are 5 defective bulb.

Ans

Given: 'P' probability of finding defect

$$P = 3\%$$

'n' total no: of sample

$$n = 100$$

IF X is the no: of defective bulb:
To find $P(X=3)$

$$X \sim P(X=3)$$

$$P(X=3) = \frac{e^{-3} 3^n}{n!}$$

$$P(X=5) = \frac{e^{-3} 3^5}{5!}$$

$$= \underline{\underline{0.1008188134}}$$

Q.

If the probability that an individual suffers a bad reaction from a certain injection is 0.002. Determine the probability that out of 1000 individual

1: Exactly 3 will suffer a bad reaction

2: More than 2 will suffer a bad reaction.

Ans:

Given 'P' = 0.002.

$$n = 1000$$

X = individual suffering a bad reaction

$$\lambda = n \times p = 1000 \times 0.002$$

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(x=3) = e^{-\lambda} \frac{\lambda^3}{3!}$$

$$\lambda = n \times p = 1000 \times 0.002 = 2$$

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \frac{e^{-2} 2^3}{3!}$$

$$\text{To find } P(x=3) = 0.1804470443$$

$$\text{To find } P(x \geq 3) = 1 - P(x < 3)$$

$$= 1 - (P(x=0) + P(x=1) + P(x=2))$$

$$= 1 - ((0.1353352832) + (0.2706705665) + (0.2706705665))$$

$$= 0.3233235838$$